Exception Handling

- **Exception** is a runtime error that occurs during program execution.
- Exception is caused due to errors that **cannot be detected during compile time.**
- ❗ **An exception if not handled will make the program crash.**
- Various exception classes are provided by Python library to handle different types of exceptions.
- All classes are subclasses of the base class called **Exception.**

Keywords used in handling exceptions are:

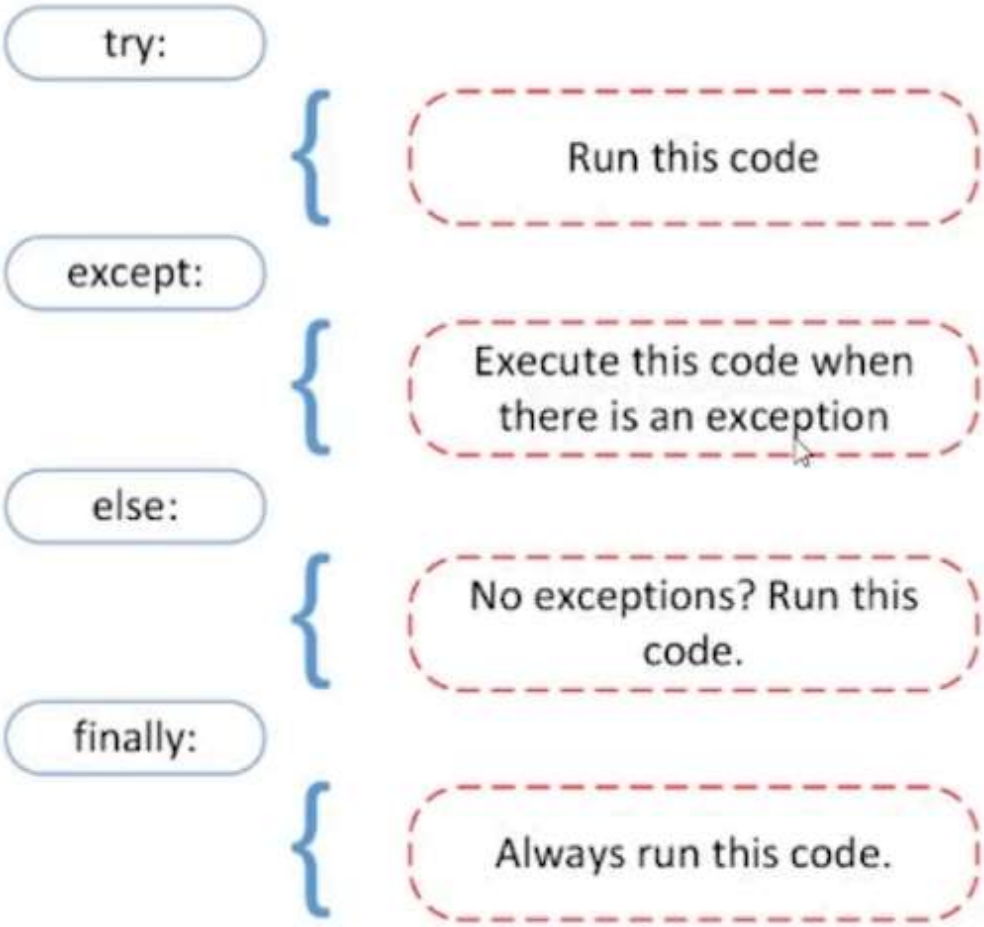
- `try`
- `except`

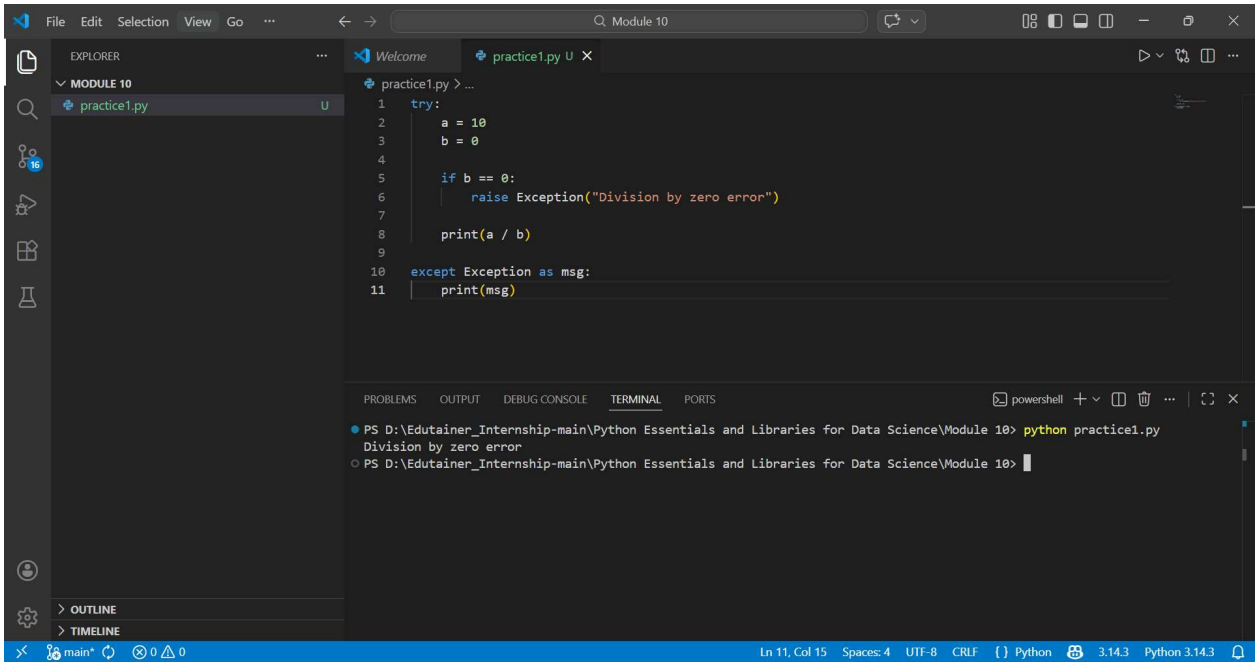
Exception handling in C++:-

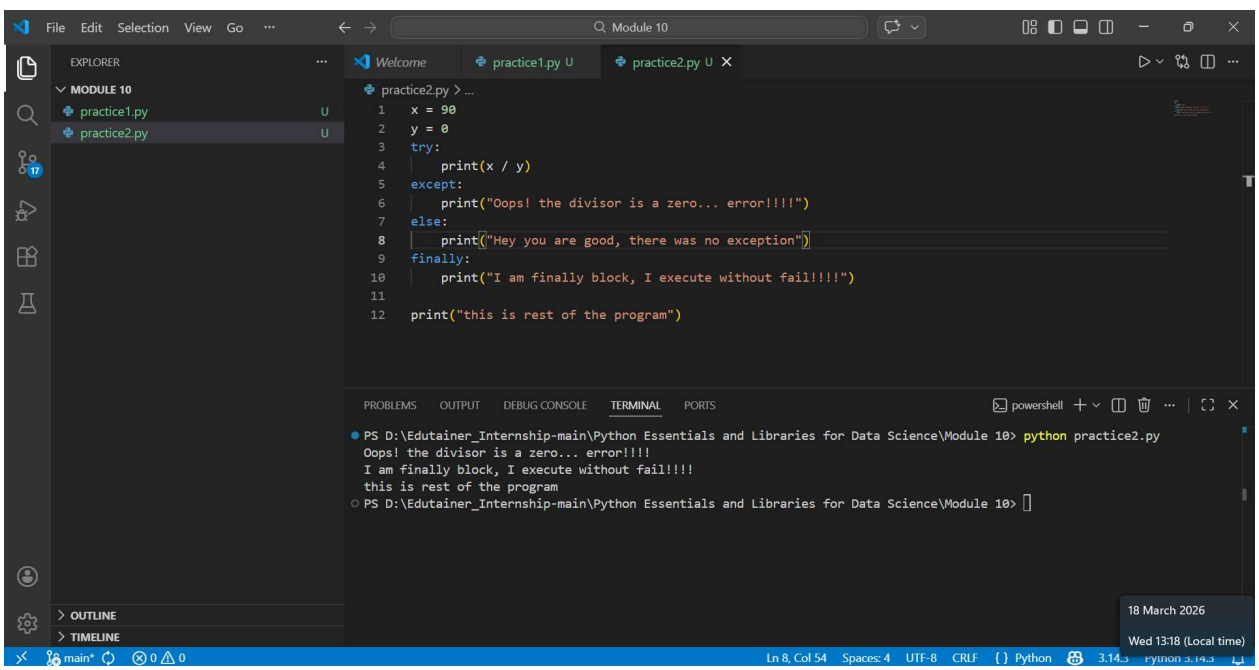
User can create **custom exception classes** by deriving classes from the **Exception base class.**

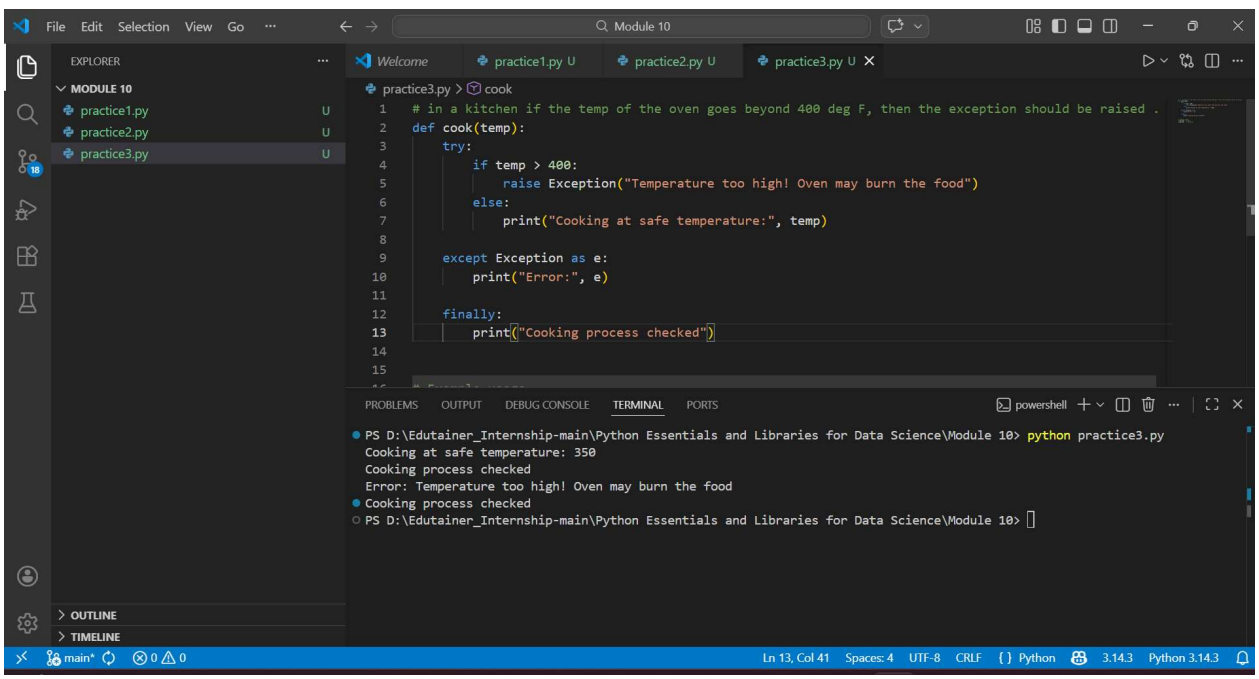
The main advantage of handling an exception is that it ensures the program **does not crash** and continues with the next normal execution flow.

However, the **damage done due to exception cannot be salvaged.**


try:
Run this code
except:
Execute this code when there is an exception
else:
No exceptions? Run this code.
finally:
Always run this code.






```python
# in a kitchen if the temp of the oven goes beyond 400 deg F, then the exception should be raised .
def cook(temp):
    try:
        if temp > 400:
            raise Exception("Temperature too high! Oven may burn the food")
```

```python
        else:
            print("Cooking at safe temperature:", temp)

    except Exception as e:
        print("Error:", e)

    finally:
        print("Cooking process checked")


# Example usage
cook(350)    # Safe
cook(450)    # Exception
```